Chapter 1

TT

1.1 CIC System

We first introduce some symbols in order to construct a formal language:

- $V = \{v_0, v_1, \cdots\}$ is the set of term variables;
- $A = \{\alpha_0, \alpha_1, \cdots\}$ is the set of type variables (below we take the union of A and V, and denote it by V);
- $C = \{\text{all UTF-8 strings}\}$ is the set of constants that are new definitions;
- $I = \{\text{Ind}, \text{Cstr}, \text{Elim}\}$ is the set of symbols used in defining inductions;
- $S = \{\text{Set}, \text{Prop}\} \cup \{\text{Type}_i\}_{i \geq 0}$ is the set of sorts;
- $P = \{., :, \equiv, (,)\}$ is the set of punctuation;
- $B = \{\lambda, \Pi\}$ is the set of binders.

Then we let
$$\mathrm{F} = C|V|S|(BV : \mathrm{F}.\mathrm{F})|(I\mathrm{F})|(\mathrm{FF})$$

Axioms $\mathcal{A} \subset S^2$ and rules $\mathcal{R} \subset S^3$ are defined as follows

$$\begin{gathered}\mathcal{A} = \{(\text{Set}, \text{Type}_0), (\text{Prop}, \text{Type}_0), (\text{Type}_i, \text{Type}_{i+1})\} \\ \mathcal{R} = \{(\text{Prop}, \text{Prop}, \text{Prop}), (\text{Set}, \text{Prop}, \text{Prop}), (\text{Type}_i, \text{Prop}, \text{Prop}), \\ (\text{Prop}, \text{Set}, \text{Set}), (\text{Set}, \text{Set}, \text{Set}), (\text{Type}_i, \text{Type}_j, \text{Type}_{\max\{i,j\}})\}.\end{gathered}$$

Note that $\Pi(\boldsymbol{x} : \boldsymbol{A})\varphi$ is the abbreviation of $\Pi x_0 : A_0.\Pi x_1 : A_1 \cdots \Pi x_m : A_m.\varphi$. $\boldsymbol{x} : \boldsymbol{A}$ is the abbreviation of $x_0 : A_0 x_1 : A_1 \cdots x_m : A_m$.

Definition 1.1.1 *The Arity has the form*

$$\mathrm{A} = S|(\Pi V : \mathrm{F}.\mathrm{A})$$

For any $X \in C$, the Strictly Positive Types has the form

$$\mathrm{P}_X = X|(\mathrm{P}_X \underbrace{\mathrm{F}}_{X \notin \mathrm{F}})|(\Pi V : \underbrace{\mathrm{F}}_{X \notin \mathrm{F}}.\mathrm{P}_X)$$

The Constructors has the form

$$\mathrm{C}_X = X|(\mathrm{C}_X \underbrace{\mathrm{F}}_{X \notin \mathrm{F}})|(\Pi V : \underbrace{\mathrm{F}}_{X \notin \mathrm{F}}.\mathrm{C}_X)|(\Pi V : \mathrm{P}_X.\underbrace{\mathrm{C}_X}_{V \notin \mathrm{C}_X})$$

Any element $\mathrm{C}_X \smallsetminus \mathrm{P}_X$ is called recursive.

Definition 1.1.2 *A definition D has the form*

$$\boldsymbol{x} : \boldsymbol{A} \quad c\boldsymbol{x} \equiv d(\boldsymbol{x}) : a$$

note that $d(\boldsymbol{x})$ refers $d \in \mathrm{F}$ and $\mathrm{Fr}\, d \subset \boldsymbol{x}$, that $x_i \in V, c \in C$ and that all A_i, d, a are in F.

A environment Δ has the form $D_0 D_1 \cdots D_n$.

Definition 1.1.3 *A judgment has the form*

$$\Delta \quad \boldsymbol{x} : \boldsymbol{A} \quad \varphi : a$$

Definition 1.1.4 *Given $\Delta = D_0 D_1 \cdots D_n$, $R_\Delta \subset \mathrm{F}^2$ is the smallest relation satisfies the following properties:*

- ⋄ *contains $\{(c\boldsymbol{U}, d(\boldsymbol{U})) : \boldsymbol{x} : \boldsymbol{A}\ c\boldsymbol{x} \equiv d(\boldsymbol{x}) : a \in D \in \Delta$ and $\boldsymbol{U} \in \mathrm{F}^*\}$;*
- ⋄ *closed under the following operations: $\mathrm{F}-, -\mathrm{F}, \lambda V : \mathrm{F}.-, C\boldsymbol{p}_1 - \boldsymbol{p}_2$.*

Definition 1.1.5 *R_β is the smallest relation satisfies*

- ⋄ *contains $\{((\lambda x : a.\varphi)\chi, \varphi\frac{\chi}{x}) : x \in V$ and all $\varphi, \chi, a \in \mathrm{F}\}$;*
- ⋄ *closed under the following operations: $\mathrm{F}-, -\mathrm{F}, BV : \mathrm{F}.-, BV : -.\mathrm{F}, C\boldsymbol{p}_1 - \boldsymbol{p}_2$.*

Proposition 1.1.6 *Any element in C_X and P_X has the form*

$$\Pi(\boldsymbol{x} : \boldsymbol{M}).(X\boldsymbol{N})$$

which is unique ***under*** *$\sim_{\beta,\Delta}$.*

Suppose X is a constant symbol, $c_i \in \mathrm{C}_X$, and $\Pi(\boldsymbol{x} : \boldsymbol{A}).s \in \mathrm{A}$, then

$$\begin{array}{llll}
\Delta & \Gamma & X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s & c_0 : s \\
 & & \vdots & \\
\Delta & \Gamma & X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s & c_m : s \\
\Delta & \Gamma & \mathrm{Ind}X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s &
\end{array}$$

and it immediately follows that

$$\begin{array}{lll}
\Delta & \Gamma & \mathrm{Ind}X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s \\
\Delta & \Gamma & \mathrm{CstrInd}Xi : c_i\frac{\mathrm{Ind}X}{X}
\end{array}$$

(i is a constant belongs to C)

Assume that

- ⋄ $\mathrm{F} \ni X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s$
- ⋄ $C \ni Y : \Pi(\boldsymbol{x} : \boldsymbol{A}).s'$
- ⋄ $c \in \mathrm{C}_X$
- ⋄ $P \in \mathrm{P}_X$
- ⋄ $f_0, \cdots, f_m \in C$
- ⋄ $\boldsymbol{a} \in \mathrm{F}^*$

we can define a new function $E_{X,Y} : \mathrm{C}_X \to \mathrm{C}_Y$ inductivly, as follows

$$\begin{aligned}
&E_{X,Y}(X) = X \\
&E_{X,Y}(cM) = E_{X,Y}(c)M \\
&E_{X,Y}(P \to c) = P \to P[Y/X] \to E_{X,Y}(c) \\
&E_{X,Y}(\Pi x : M.c) = \Pi x : M.E_{X,Y}(c)
\end{aligned}$$

and note that $E_{\Phi,Y}(c) := E_{X,Y}(c)\frac{\Phi}{X}$

In addition, we have

$$\begin{array}{lll} \Delta & \Gamma & d : (\mathrm{Ind}X)\boldsymbol{a} \\ \Delta & \Gamma & Y : \Pi(\boldsymbol{x} : \boldsymbol{A}).s' \\ \Delta & \Gamma & f_0 : E_{\mathrm{Ind}X,Y}(c_0) \\ & & \vdots \\ \Delta & \Gamma & f_m : E_{\mathrm{Ind}X,Y}(c_m) \\ \Delta & \Gamma & \mathrm{Elim}dYf_0f_1\cdots f_m : Y\boldsymbol{a} \end{array}$$

where $I\boldsymbol{a}$ is defined inductivly, that is, $Ia_1a_2\cdots a_n = (Ia_1)(a_2\cdots a_n)$.

Hypothesis: $d : \mathrm{Ind}X\boldsymbol{a} \leftrightarrow d : c \in \mathrm{C}_{\mathrm{Ind}X}$

Assume that

- $\diamond$ $\mathrm{F} \ni X : \Pi(\boldsymbol{x} : \boldsymbol{A}).s$
- $\diamond$ $C \ni Y : \Pi(\boldsymbol{x} : \boldsymbol{A}).(X\boldsymbol{x}) \to s'$
- $\diamond$ $\mathrm{F} \ni d : c \in \mathrm{C}_X$
- $\diamond$ $P = \Pi(\boldsymbol{x}' : \boldsymbol{P}').(X\boldsymbol{m}') \in \mathrm{P}_X$
- $\diamond$ $p \in V$
- $\diamond$ $f_0, \cdots, f_m \in C$
- $\diamond$ $\boldsymbol{a} \in \mathrm{F}^*$

we can define a new function $E_{X,Y,d} : \mathrm{C}_X \to \mathrm{C}_Y$ inductivly, as follows

$$\begin{aligned} &E_{X,Y,d}(X) = Y \\ &E_{X,Y,d}(cM) = (E_{X,Y,d}(c)M)d \\ &E_{X,Y,d}(P \to c) = \Pi p : P.(\Pi(\boldsymbol{x}' : \boldsymbol{P}').Y\boldsymbol{m}'(p\boldsymbol{x}') \to E_{X,Y,dp}(c)) \\ &E_{X,Y,d}(\Pi x : M.c) = \Pi x : M.E_{X,Y,dx}(c) \end{aligned}$$

we have

$$\begin{array}{lll} \Delta & \Gamma & d : (\mathrm{Ind}X)\boldsymbol{a} \\ \Delta & \Gamma & Y : \Pi(\boldsymbol{x} : \boldsymbol{A}).((\mathrm{Ind}X\boldsymbol{x}) \to s') \\ \Delta & \Gamma & f_0 : E_{\mathrm{Ind}X,Y,\mathrm{CstrInd}X0}(c_0) \\ & & \vdots \\ \Delta & \Gamma & f_m : E_{\mathrm{Ind}X,Y,\mathrm{CstrInd}Xm}(c_m) \\ \Delta & \Gamma & \mathrm{Elim}dYf_0f_1\cdots f_m : (Y\boldsymbol{a}d) \end{array}$$

Again, suppose $u, v \in \mathrm{F}$ we define a function $H_{X,u,v} : \mathrm{C}_X \to \mathrm{F}$, by letting

$$\begin{aligned} &H_{X,u,v}(X) = v \\ &H_{X,u,v}(cM) = H_{X,u,v}(c) \\ &H_{X,u,v}(P \to c) = \lambda p : P.H_{X,u,(vp\lambda(\boldsymbol{x}:\boldsymbol{P}).(v\boldsymbol{m}'(p\boldsymbol{x})))}(c) \\ &H_{X,u,v}(\Pi x : M.c) = \lambda x : M.H_{X,u,vx}(c) \end{aligned}$$

Definition 1.1.7 *Let* $\boldsymbol{f}$ *be* $f_0f_2\cdots f_m$, $\to_\iota$ *is a relation in* F^2 *contains*

$$\mathrm{Elim}(\mathrm{CstrInd}Xi)\boldsymbol{m}'Y\boldsymbol{f} \to_\iota H_{\mathrm{Ind}X,\lambda(\boldsymbol{x}:\boldsymbol{A}).\lambda d:\mathrm{Ind}X\boldsymbol{x}.\mathrm{Elim}GY\boldsymbol{f},f_i}(c_i)\boldsymbol{m}'$$

In COC, we have introduced the following sequent rules.

(Sort)

If $(s_1, s_2) \in \mathcal{A}$, then

$$\_ \quad \_ \quad s_1 : s_2$$

(Var)

If $x \in V \smallsetminus \boldsymbol{x}$, then

$$\begin{array}{lll} \Delta & \Gamma & a : s \\ \Delta & \Gamma & x : a \quad x : a \end{array}$$

(Weak)

If $x \in V \smallsetminus \boldsymbol{x}$, then

$$\begin{array}{lll} \Delta & \Gamma & \varphi : a \\ \Delta & \Gamma & b : s \\ \Delta & \Gamma & x : b \quad \varphi : a \end{array}$$

(Form)

If $(s_1, s_2, s_3) \in \mathcal{R}$, then

$$\begin{array}{llll} \Delta & \Gamma & a_1 : s_1 & \\ \Delta & \Gamma & x : a_1 & a_2 : s_2 \\ \Delta & \Gamma & \Pi x : a_1.a_2 : s_3 & \end{array}$$

(Appl)

$$\begin{array}{lll} \Delta & \Gamma & \varphi : \Pi x : a.b \\ \Delta & \Gamma & \chi : a \\ \Delta & \Gamma & \varphi\chi : b\frac{\chi}{x} \end{array}$$

(Abst)

$$\begin{array}{llll} \Delta & \Gamma & x : a & \varphi : b \\ \Delta & \Gamma & \Pi x : a.b : s & \\ \Delta & \Gamma & (\lambda x : a.\varphi) : (\Pi x : a.b) & \end{array}$$

(Def)

$$\begin{array}{lll} \Delta & \Gamma & \varphi : a \\ \Delta & \boldsymbol{y} : \boldsymbol{B} & d(\boldsymbol{y}) : b \\ \Delta D & \Gamma & \varphi : a \end{array}$$

where D is the string

$$\boldsymbol{y} : \boldsymbol{B} \quad c\boldsymbol{y} \equiv d(\boldsymbol{y}) : b$$

that $d(\boldsymbol{y})$ could be an empty word.

(Inst)

If Δ contains the element

$$\boldsymbol{y} : \boldsymbol{B} \quad c\boldsymbol{y} \equiv d(\boldsymbol{y}) : a$$

that $d(\boldsymbol{y})$ could be empty, and we have $\boldsymbol{U} \in \mathrm{F}^*$ with the same size of $\boldsymbol{y}$, then

$$\begin{array}{lll} \Delta & \Gamma & s_1 : s_2 \ (\in \mathcal{A}) \\ \Delta & \Gamma & U_0 : B_0\frac{\boldsymbol{U}}{\boldsymbol{y}} \\ & \vdots & \\ \Delta & \Gamma & U_m : B_m\frac{\boldsymbol{U}}{\boldsymbol{y}} \\ \Delta & \Gamma & c\boldsymbol{U} : a\frac{\boldsymbol{U}}{\boldsymbol{y}} \end{array}$$

(Conv)

If $a \sim_{\Delta,\beta} a'$, then

$$\begin{array}{lll} \Delta & \Gamma & \varphi : a \\ \Delta & \Gamma & a' : s \\ \Delta & \Gamma & \varphi : a' \end{array}$$